

Process Control: Level Control

Pre-Lab Plan
Unit Operations Lab #6
November 14, 2019

Group #2, Thursday, PM section

Diana Awimrin
Leena Daniel
Alice Guzik
Esteban Vazquez- Elizondo

1. Experimental Objective

The overall objective is to learn the fundamentals of process control by controlling levels of water in a process vessel and to determine the effects of different ranges of inflow and outflow to/from the process vessel.

2. Experimental

2.1 Apparatus

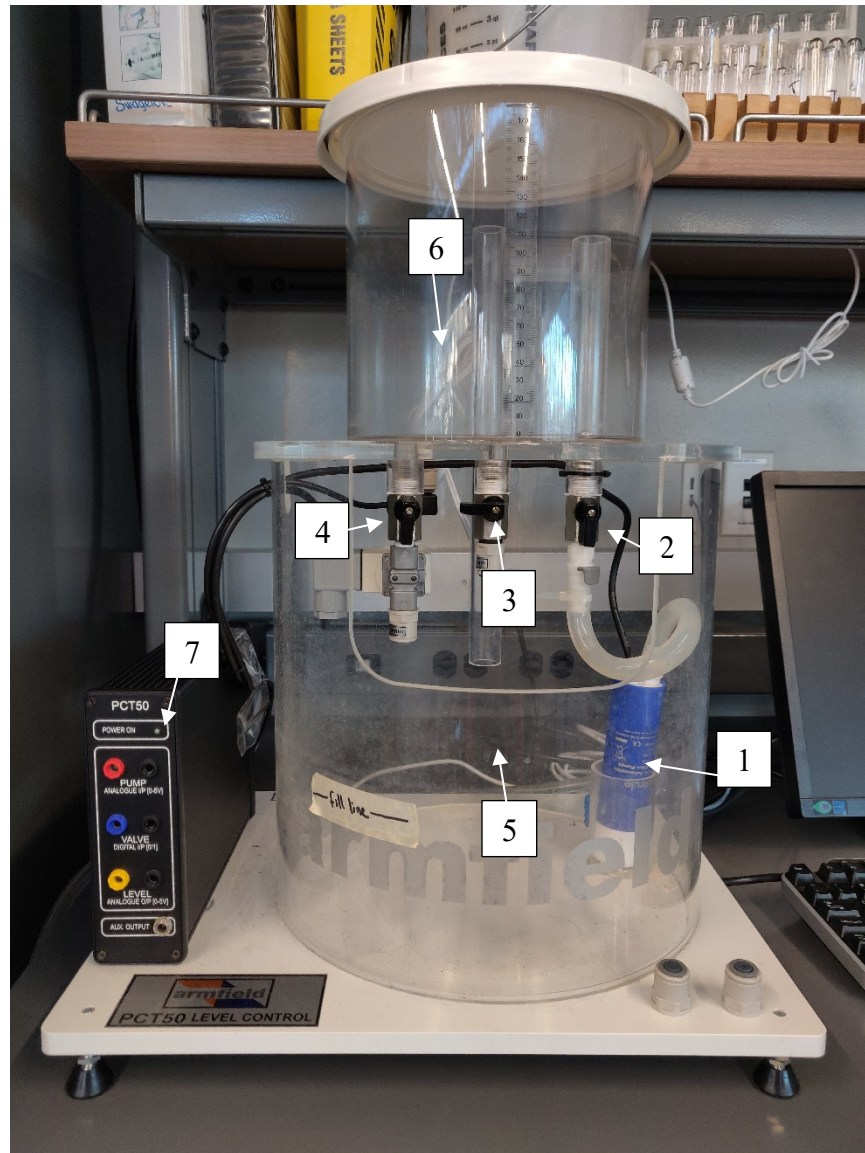


Figure 1: Apparatus of the Sump Tank

<u>Number</u>	<u>Materials</u>	<u>Brief Description</u>
1	Pump	Moves water from sump tank to the process tank
2	Inlet Valve	Allows the water to enter the process tank
3	Outlet Valve	Let's the water flow from the process tank back into the sump tank
4	Solenoid Valve	Introduces disturbances to the process tank
5	Sump Tank	Main tank that holds water that is supplied to the process tank
6	Process Tank	Second water vessel where water level is maintained
7	Controller	Transmits signals based on water level into the computer

2.2 Materials and Supplies

<u>Materials</u>	<u>Brief Description</u>
Water	Material being pumped into the vessel
Bucket	To carry the water and fill the sump tank
Computer	Used to display and read data collected

2.3 Experimental Plan

Initial Filling of the Sump Tank

1. Make sure that the drain valve is closed
2. Fill the sump tank up to a water level of approximately 30mm below the cut for normal use, using a water pump.

Equipment Setup

3. Make sure the power supply is connected to the socket marked 24V
4. Fully open the outlet valve CV2 and add a orifice, with a diameter of 4mm, below it and add a 2 mm orifice under the solenoid valve
5. Make sure the PC is switched on and the PCT50 software is switched on and make sure the Virtual COM port is enabled using the red telephone icon
6. Fully open the outlet valve CV2 and the inlet valve CV1 and then display a mimic diagram of the process
7. Turn the Power on and make sure the reading is zero in the empty vessel
8. Set the pump speed to 50% to allow water into the process vessel

9. Adjust inlet valve CV1 until the level is steady and the flow into the vessel matches the flow out of the vessel
10. Reset the pump to 0% and allow the vessel to drain the outlet valve CV2

Experiment A

11. Observe the rise of levels as the speed of the pump is increased and the flow into the vessel is greater than the flow out
12. Adjust the speed to be the same as the outflow to get a constant level
13. Choose the “GO” icon and record the level reading L1 in mm
14. Open the solenoid valve and observe the levels decreasing
15. Once level has dropped 20mm, record the current pump speed and then increase the pump speed to get the original level
16. Close the solenoid valve button
17. As the level rises, record the pump speed after the level has risen 20mm and then reduce the pump speed to original level
18. Click the “STOP” icon and then display the graph of the results
19. Reduce the pump speed and drain the process vessel
20. Create a new sheet
21. Increase the pump speed until the level starts to rise and allow the speed to be constant as the vessel fills up to 150mm
22. Stop the pump and drain the vessel
23. Start the pump at 75% speed and let the vessel fill up until it's at 150mm and then stop the pump and drain the vessel
24. Repeat exercise with different sizes of orifice fitted to the outlet valve CV2 and the solenoid valve
25. Save data onto a USB

Experiment C: P Controller

26. Choose the PID box and set the Proportional Band P to 200%, Integral Time I to 0, Derivation time D to 0, and set the Set Point to 75mm
27. Choose the automatic mode of operation and begin logging the data
28. Once the levels become steady, open the solenoid valve to disturb the process
29. Once the level becomes steady, close the solenoid valve and allow the level to settle again
30. Change the Set Point to 100mm and once it settles, return the set point to 75mm
31. Change the P term to 100% in the PID controller
32. Disturb the process by opening and closing the solenoid valve
33. Change the P term to 50% and repeat step 31
34. Change the P term to 20% and repeat step 31
35. Change the P term to 10% and repeat step 31
36. Change the P term to 5% and repeat step 31
37. Change the P term to 2% and repeat step 31
38. Finish the data logging

Experiment C: P +I Controller

39. Create a new sheet and repeat step 26 and start recording the data

40. Choose the automatic mode and let the water level become stabilized
41. Change I term to 100 seconds and allow level to settle
42. Open the solenoid valve to disturb the process
43. Once the level settles, close the solenoid valve and let the level stabilize
44. Change the Set Point to 100 mm and after the level settles, change it to 75mm
45. Adjust the I term to 50 seconds and then when the level is settles at the set point, open and close the solenoid valve
46. Adjust the set point to 100 mm and once the level settles, change the set point to 75mm
47. Reduce the I term and disturb the process by opening and closing the solenoid valve
48. Change Set Point to 100mm and then after the level stabilizes, change it to 75 mm
49. Continue steps 47-48 until the levels are unstable

Project Safety Evaluation

Project Title: _____ Process Control: Level Control

Date: _____ November 14, 2019

	<i>Yes/No</i>	
Potential electrical hazards?	Yes	If yes, identify conditions: Watch out for exposed wires near the water.
Potential mechanical hazards?	Yes	If yes, identify conditions: The pump can cause a hazard.
Potential pressure hazards?	No	If yes, identify conditions:
Potential temperature hazards?	No	If yes, identify conditions:
Hazardous/toxic chemicals involved?	No	If yes, hazards involved:
Gloves required?	No	If yes, specify type:
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Water
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	25°C	Ambient Temperature
Pressure (psia)	1 atm	Atmospheric Pressure
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
	Yes	Drain the Unit using the ball valve located at the rear and shut off the system

I have read relevant background material for the Unit Operations Laboratory entitled: “_____ Process Control: Level Control_____” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

_____ 11/14/19
signature of team member date

_____ 11/14/19
signature of team member date

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signature of team member date

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signature of team member date